IPB University | Tri-U 32

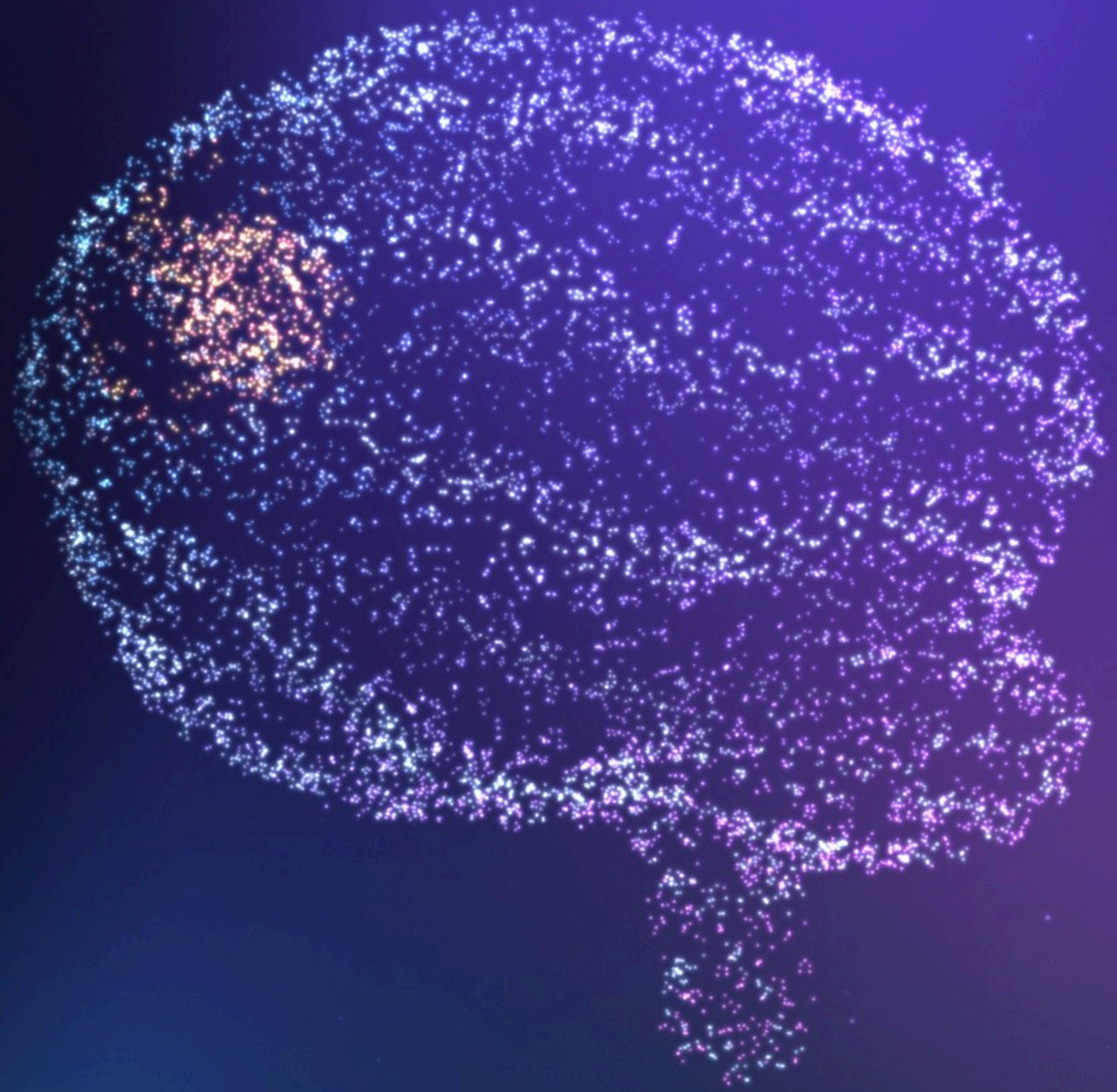
Tri-U International Joint Seminar & Symposium 2026 · Chiang Mai University, Thailand

[Healthcare Innovation]

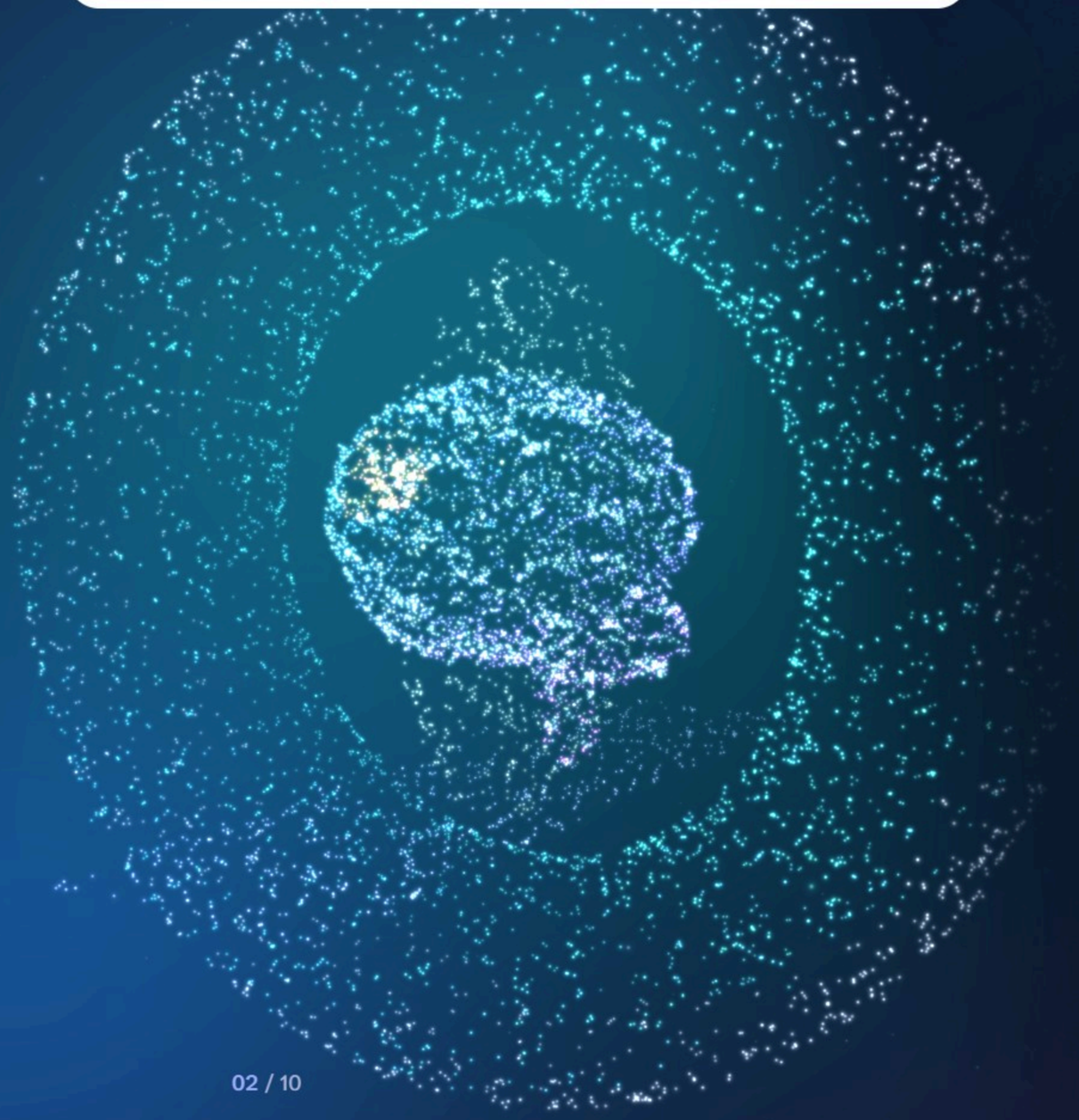
An exploratory multi-scale analysis linking MRI radiomics with KDR/VEGFR-2 expression and kinase-domain structure in glioblastoma

Muhammad Dandy Nugraha · Wisnu Ananta Kusuma
Department of Bioinformatics, IPB University

15 months median survival, glioblastoma



Background: illustrative animation, not data.



[The question]

Measure KDR from an MRI, not from tissue.

Bevacizumab

Blocks the signal that grows new blood vessels into the tumour.

VEGFR-2 · KDR

The receiver, and the gene that builds it.

91 patients · all public data

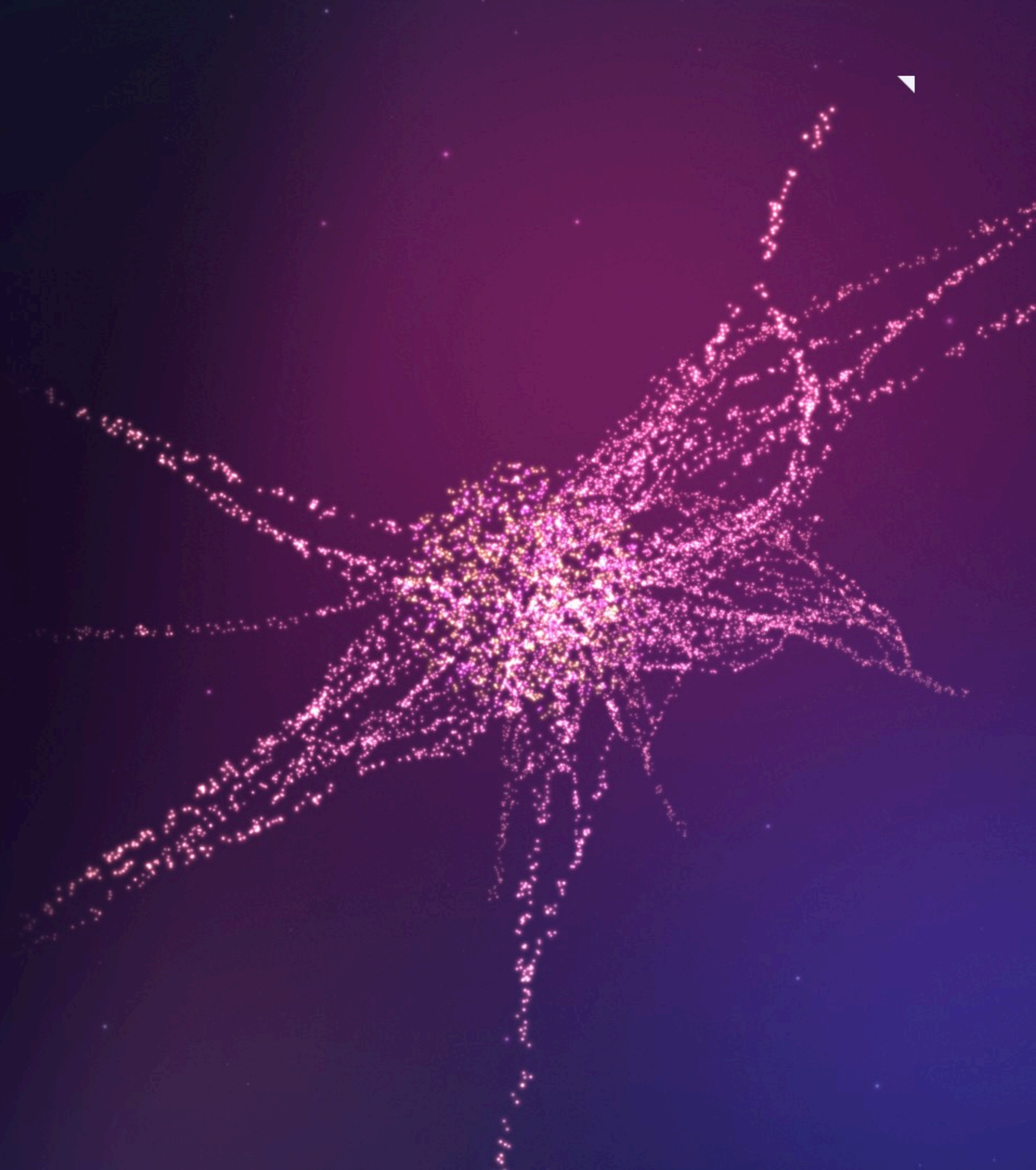
[Background]

A signal does nothing unless something receives it.

01 A tumour grows faster than its own blood supply.

02 It sends a signal for new blood vessels: angiogenesis.

03 More receivers means more blood vessels, and more for the drug to block.





[Methods]

Image, gene, structure.

01 Image
PyRadiomics turns each MRI into measurements we can name.

02 Gene
Does it rise and fall together with KDR in the same patient?

03 Structure
A pocket where a drug can attach. VEGFR-2 kinase domain, PDB 2OH4.

Why not deep learning? Ninety one patients is far too few.

[Two checks]

Two checks. Neither was in my first draft.

01

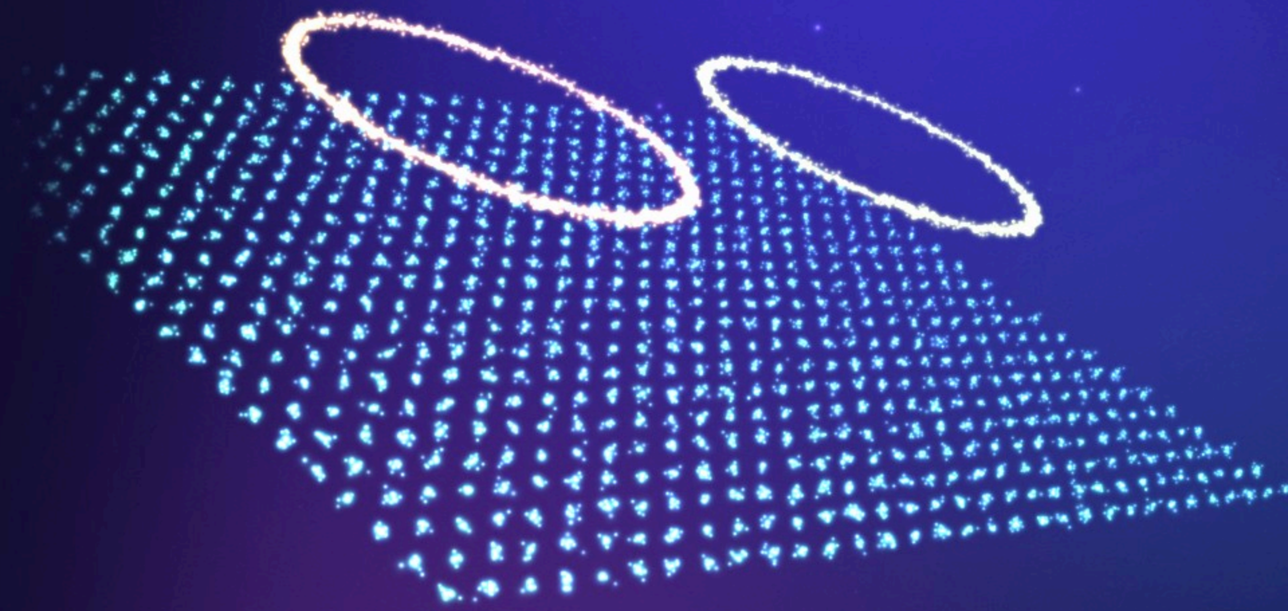
Multiplicity

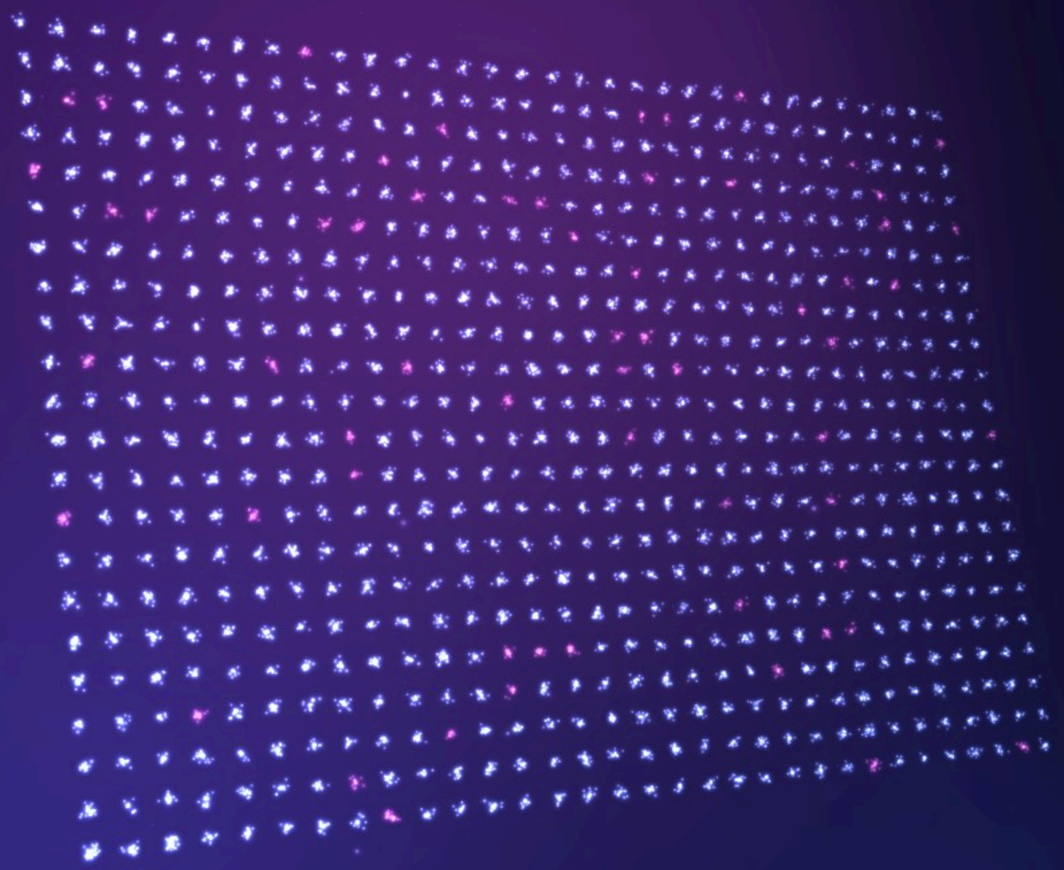
Running a very large number of tests at the same time.

02

Leakage

Marking your own work fairly.





[Check 01 · Multiplicity]

Zero survived.

704
tests, one per measurement

61
came back significant

35
expected by chance alone

0
survived FDR correction

Strongest: q = 0.530

[Check 02 · Leakage]

I used all the data.

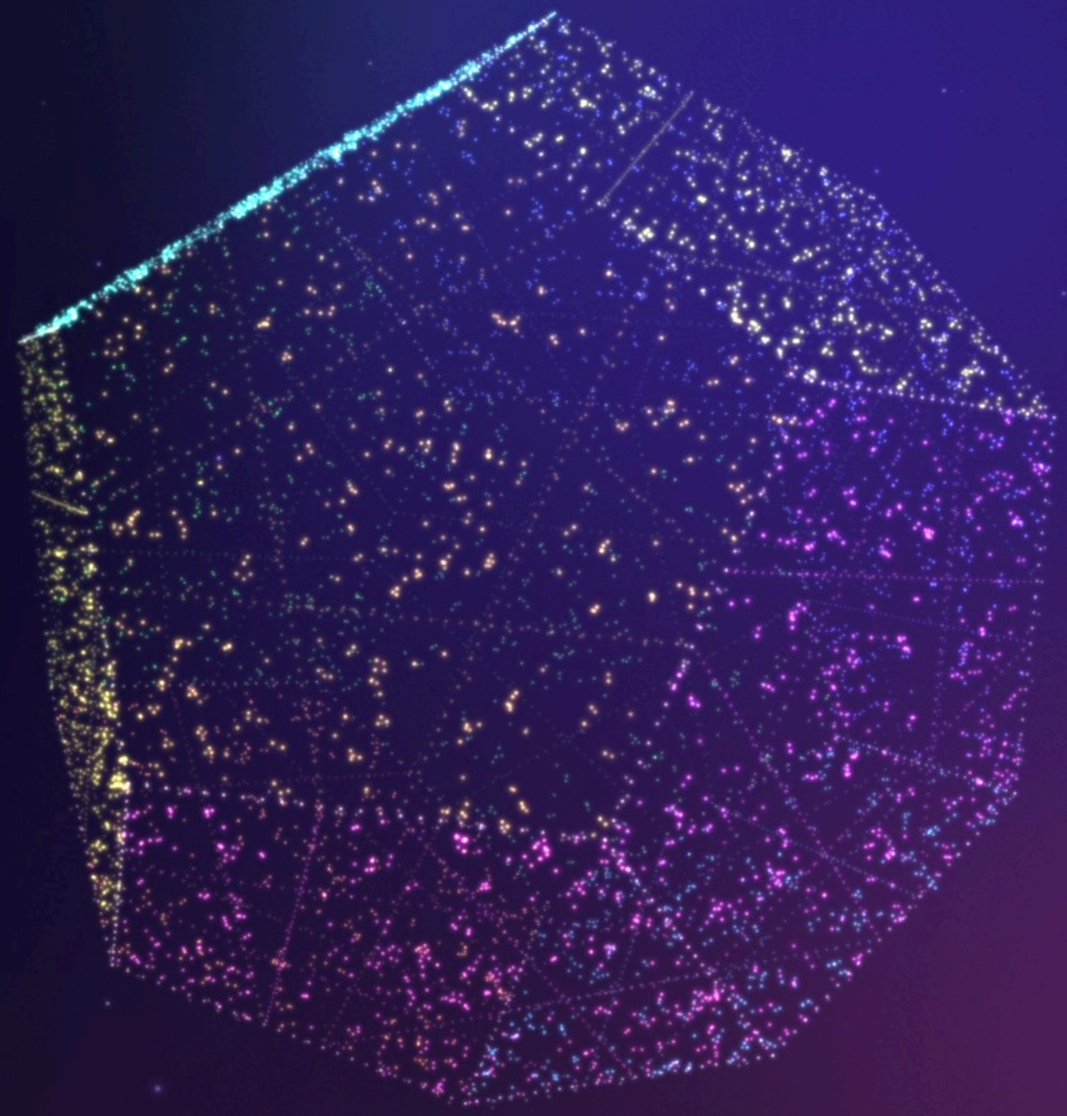
~~0.687~~ → **0.497**

AUC without nesting

nested · 0.5 means guessing

+0.190 all of it came from the mistake

Of a hundred patients, about fifty go into the wrong group.





[Provenance]

Same images. Different software. Opposite answers.

Archive · pre-computed BraTS	Ours · re-extracted PyRadiomics
0	479
survived · 91 patients	survived · 86 patients

This fits a software effect, but it does not prove one.

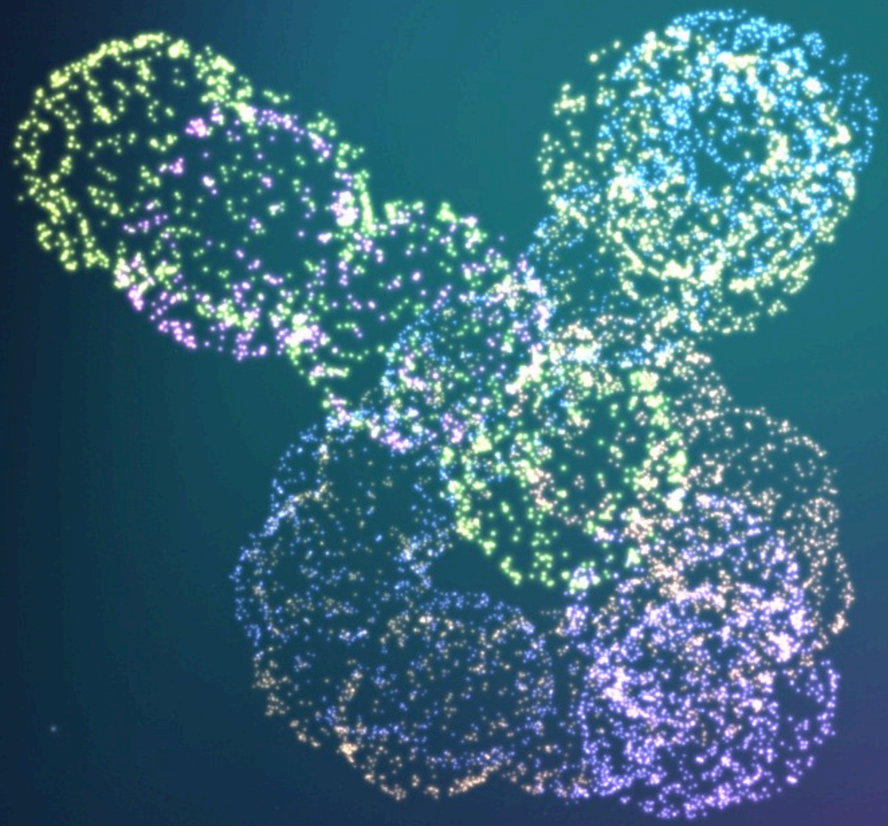
Of six genes, KDR was the strongest: q = 0.010

[Discussion]

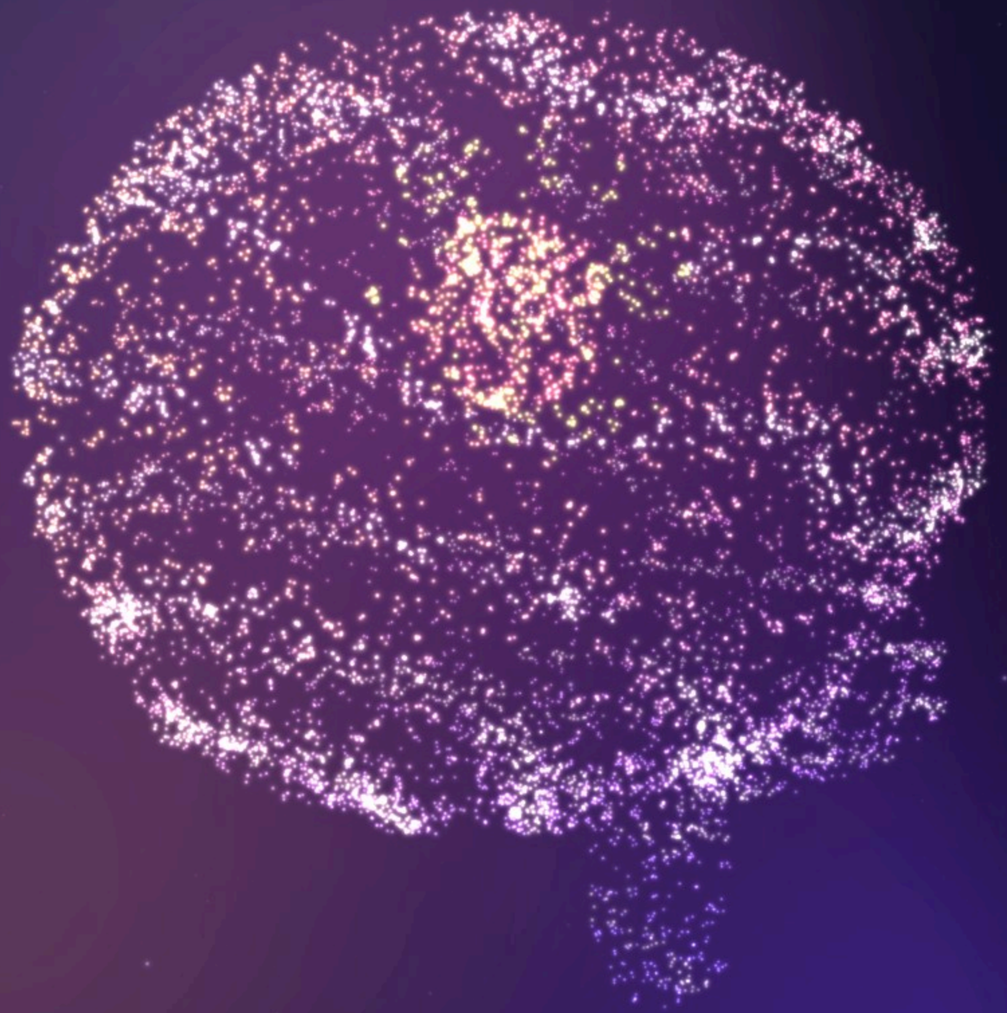
The biology did not change.
The record keeping did.

Run both checks before you publish your number.

Mine took one afternoon.



Everything here is public, the code is open.
ipb.link/triu-nugraha-2026



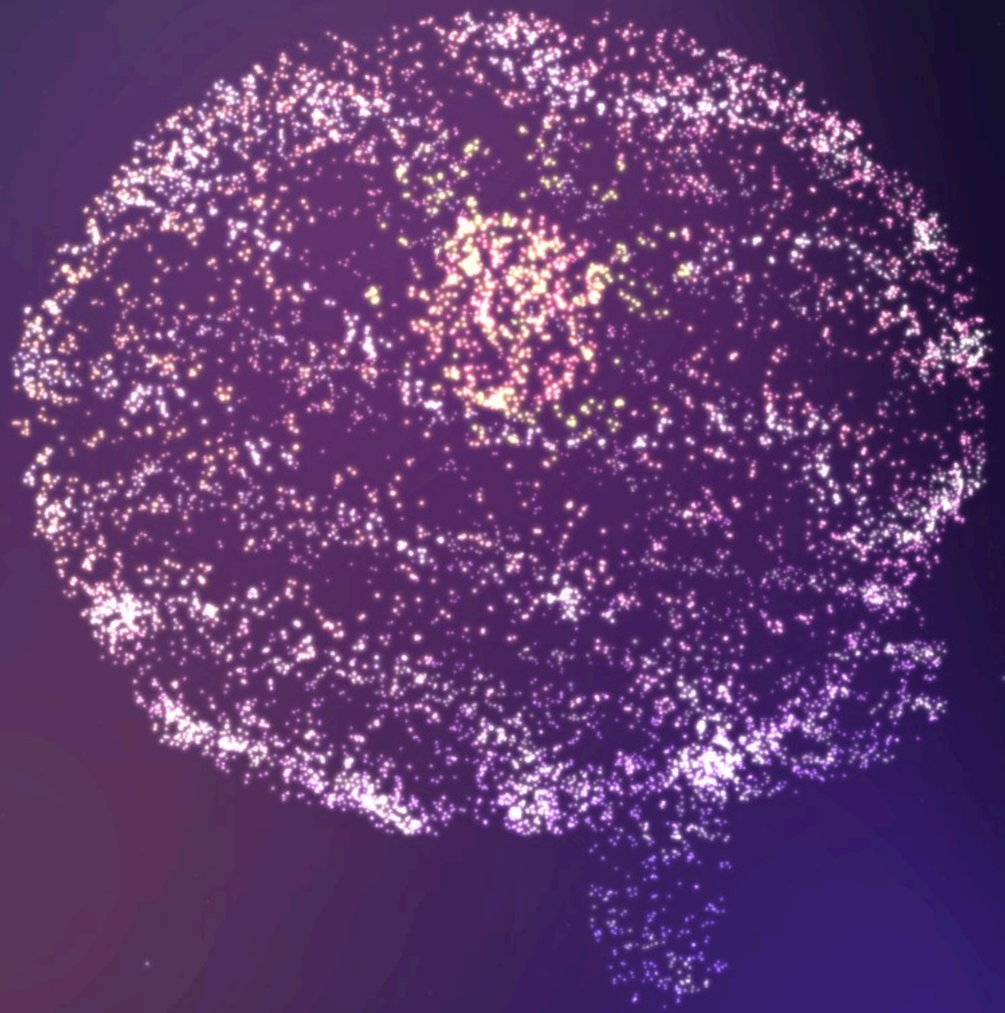
Thank you.

ขอบคุณครับ

Muhammad Dandy Nugraha · Wisnu Ananta Kusuma
Department of Bioinformatics, IPB University

muhammad_dandy@apps.ipb.ac.id
ipb.link/triu-nugraha-2026

Extra slides: validation design · software comparison · protein structure



[Backup A · Validation design]

Validation design

Non-nested

Screening and tuning fitted on the full cohort.

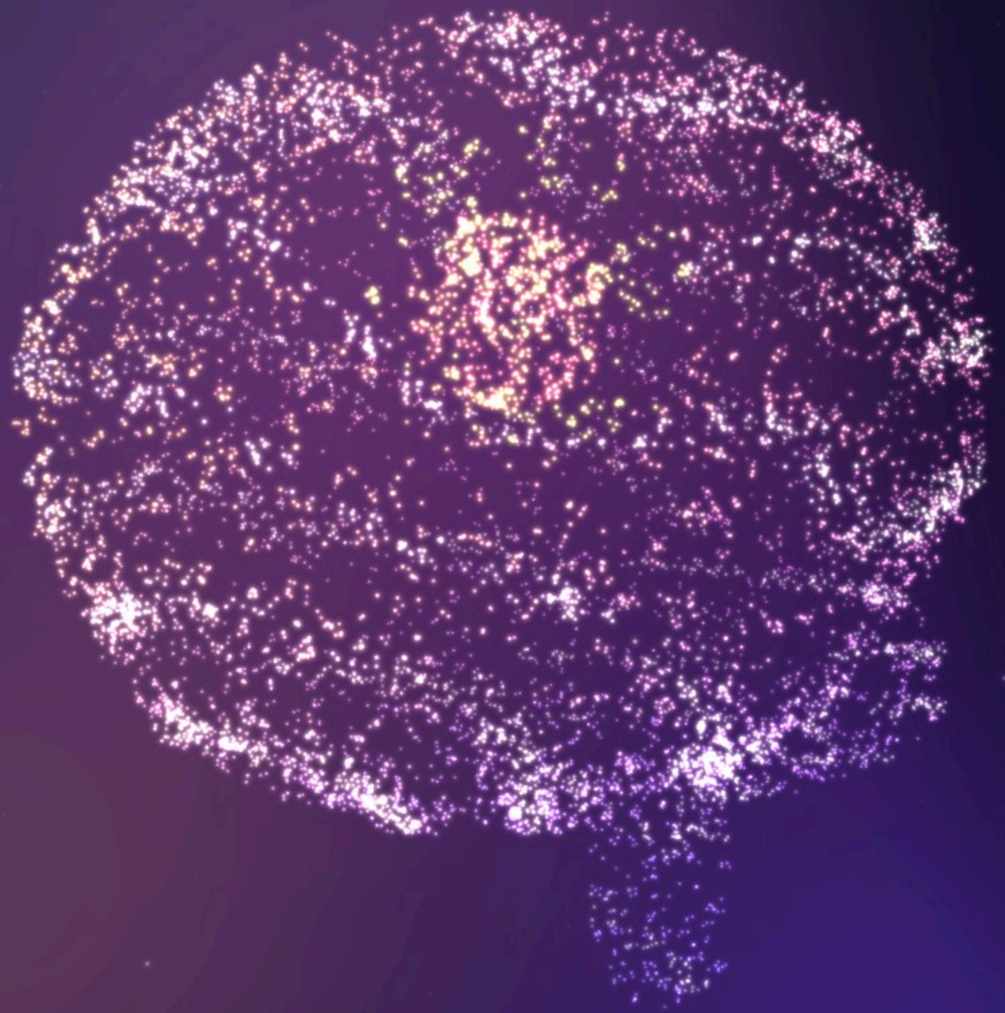
AUC 0.687 ± 0.081

Nested

Both confined to each outer training fold.

AUC 0.497 · 95% CI 0.346–0.648 · accuracy 48.4%

Median split z = −0.179 · 46 HIGH / 45 LOW



[Backup B · Software comparison]

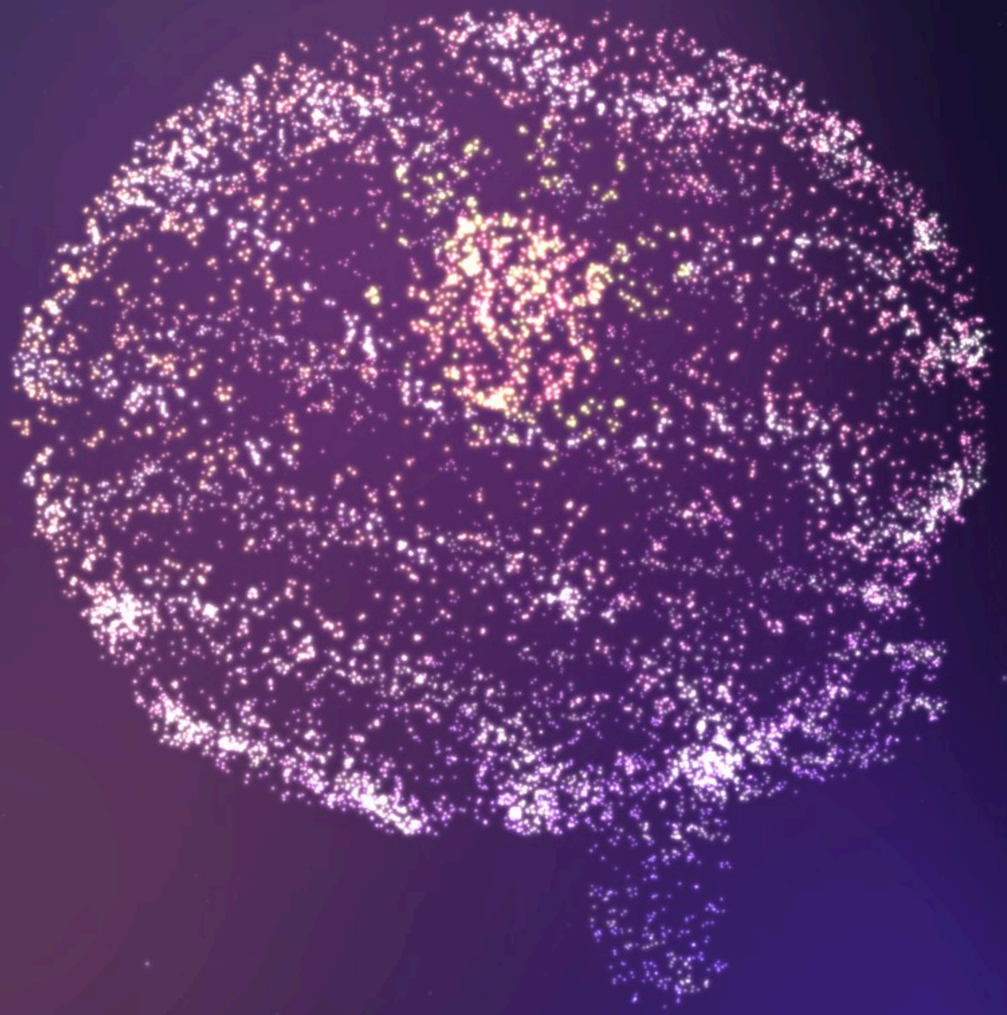
Software comparison

Table 1. Correlation with KDR expression for two descriptor sets computed from the same images.

Descriptor set	Patients	Tested	Nominal	FDR q<0.05	Min q
Pre-computed BraTS	91	704	61	0	0.530
Re-extracted PyRadiomics	86	1098	682	479	0.010

479 represents a few underlying dimensions expressed repeatedly, not 479 independent findings.

CXCL8 also survived correction (q = 0.039).



[Backup C · Protein structure]

VEGFR-2 kinase domain, PDB 2OH4.

01 Eight ATP-binding residues within 5 Å of the inhibitor.

02 Gatekeeper Val914

03 DFG motif Asp1044/Phe1045

04 Numbering offset by two from UniProt.

Qualitative visualisation supplying molecular context rather than a mechanistic bridge to the imaging model.